

Jin Hyun Park

Ph.D. Student, Computer Science and Engineering, Texas A&M University

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RESEARCH THEMES

Marks link each theme to the publications and positions below.

- ▲ **Brain-inspired neural networks.** Visual-cortex circuitry in CNNs, saliency and sparsity in neural codes. [A2, W1, A1]
- **Evolution of prediction and physical AI.** How predictive behavior emerges; learning-based control of physical systems. [J1]
- **Language models and NLP.** LLM evaluation, LLM-assisted signal correction, retrieval-augmented search. [C2, C1]

EDUCATION

Ph.D. in Computer Science and Engineering

Texas A&M University, College Station, TX, USA Advisor: Yoonsuck Choe

Aug 2022 – May 2027 (expected)

B.Sc. in Computer Science, First Class Honours

University of Bristol, Bristol, UK Advisor: Rui Ponte Costa

Sep 2018 – Jul 2021

PUBLICATIONS

C: conference, J: journal, W: workshop, A: arXiv preprint. * denotes equal contribution.

- [C2] ● Soha Lee, Soojin Lee, Heesung Yang, Hyunju Song, Hyunji Lee, Jinsan An, Jeongwan Shin, **Jin Hyun Park**, Jun Lee, Hyeyoung Park, Kilim Nam. *KoNeoBench: A Curated Evaluation Dataset for LLM Understanding of Korean Neologisms*. Findings of EMNLP, 2026.
- [C1] ● Seyyed Ali Ayati*, **Jin Hyun Park***, Yichen Cai, Marcus Botacin. *Making Acoustic Side-Channel Attacks on Noisy Keyboards Viable with LLM-Assisted Spectrograms’ “Typo” Correction*. 19th USENIX Workshop on Offensive Technologies (WOOT), 2025. [arXiv:2504.11622](https://arxiv.org/abs/2504.11622)
- [J1] ■ William Kang, Chris Anand, **Jin Hyun Park**, Yoonsuck Choe. *Evolutionary Factors Contributing to the Emergence of Prediction*. Adaptive Behavior, 2025. [doi:10.1177/10597123251364746](https://doi.org/10.1177/10597123251364746)
- [W1] ▲ Alex Alvarez*, **Jin Hyun Park***, Yoonsuck Choe. *Saliency Thresholds in Neural Code and its Relation to the Power-Law, Gaussian, and Lambert W Function*. NeurIPS Workshop on Symmetry and Geometry in Neural Representations (NeurReps), 2025. [OpenReview](#)
- [A2] ▲ **Jin Hyun Park**, Cheng Zhang, Yoonsuck Choe. *Incorporating Visual Cortical Lateral Connection Properties into CNN: Recurrent Activation and Excitatory-Inhibitory Separation*. arXiv preprint, 2025. [arXiv:2509.15460](https://arxiv.org/abs/2509.15460)
- [A1] ▲ **Jin Hyun Park**. *Representations Learnt by SGD and Adaptive Learning Rules: Conditions that Vary Sparsity and Selectivity in Neural Networks*. arXiv preprint, 2022. [arXiv:2201.11653](https://arxiv.org/abs/2201.11653)

RESEARCH EXPERIENCE

Graduate Research Assistant

Brain Networks Lab, Texas A&M University Advisor: Yoonsuck Choe

Aug 2022 – Present

- ▲ Biologically inspired neural networks: incorporated visual-cortex lateral connectivity (recurrent activation, excitatory–inhibitory separation) into CNNs [A2] and analyzed saliency thresholds in neural codes [W1].
- Evolution of prediction: studied the evolutionary factors that give rise to predictive behavior in simulated agents [J1].
- Learning-based control: developed an AI-driven smart thermostat for floor-heating systems.

Research Assistant ●

Multimodal Language and Cognition Lab, Kyungpook National University, Daegu, Korea

Jan 2022 – Jul 2022

- Research on classical NLP algorithms and large language models.
- Organized a reinforcement learning seminar series for lab members, with lecture notes and code samples.

INDUSTRY EXPERIENCE

Physical AI Ph.D. Fellow ■

Doosan Corporation, Seoul, Korea

Feb 2026 – Present

Machine Learning Engineer Intern	May – Aug 2023; Jul – Aug 2025
Machine Learning Team, Tilda Corp., Seoul, Korea	
2025: Meta-heuristic and neural methods for vehicle routing, multi-agent path finding, and time-series forecasting. 2023: Machine learning solutions based on a neural-network-guided genetic algorithm.	
Machine Learning Engineer Intern ●	Jul – Aug 2024
AI Service Part, Digital Innovation Team, Samsung Life Insurance, Seoul, Korea	
Built an automatic PDF parsing pipeline and a retrieval-augmented (RAG) search engine for insurance documents and policy retrieval.	
Machine Learning Engineer Intern	May – Jul 2024
AI Prediction Tech Team, Data Science Lab, LG Uplus, Seoul, Korea	
- Ensemble and transformer models for identifying outbound roaming travelers. - Large-scale data generation, preprocessing, and visualization with SQL and Google BigQuery.	
Business Strategy Intern	Oct – Dec 2021
Commercial Vehicle Business Innovation Team, Hyundai Motor Company, Seoul, Korea	
Short- to long-term business modeling for commercial vehicles, with a focus on FCEV and BEV.	

TEACHING AND SERVICE

Reviewer / Program Committee	
AAAI 2024, 2025, 2026; AAAI-AIA 2026; AAAI-Demo 2026; IEEE Transactions on Cognitive and Developmental Systems 2025; NeurIPS NeurReps Workshop 2026.	
Teaching Assistant	2022 – Present
CSCE 420/625 Artificial Intelligence, Texas A&M University.	
Seminar	2025
Seminar on large language models, Yonsei University.	

SELECTED PROJECTS

Depth- and Attention-Aware RGB Image Inpainting	2025
Course project, Computational Photography arXiv:2505.00735	
Dual-encoder attention architecture fusing RGB and depth features; outperforms RGB-only baselines on several metrics, with Grad-CAM confirming improved spatial reasoning.	
PEPSE: Personal Photo Search Engine	2024
Course project, Information Storage and Retrieval	
Semantic photo search combining caption generation, query embedding, and face recognition with pre-trained transformer models.	

FELLOWSHIPS, GRANTS, AND AWARDS

Physical AI Ph.D. Fellowship, Doosan Corporation	2026 – Present
Academic Excellence Grant, Texas A&M University	2025 – 2027
Travel Grants	
- Doosan Corporation, Consumer Electronics Show (CES), 2026 - Texas A&M University, NeurIPS, 2025 - Samsung Life Insurance and LG Uplus, internship travel, 2024 - Hyundai Mobis, Consumer Electronics Show (CES), 2024	
Barry Thomas Scholarship, University of Bristol	2018
Future Prize Winner Scholarship, University of Bristol	2017

SKILLS

Languages	Python, Java, C/C++, SQL
Frameworks	PyTorch, OpenCV, Google BigQuery
Areas	Deep learning, computational neuroscience, reinforcement learning, NLP and LLMs, retrieval-augmented generation, evolutionary and meta-heuristic optimization